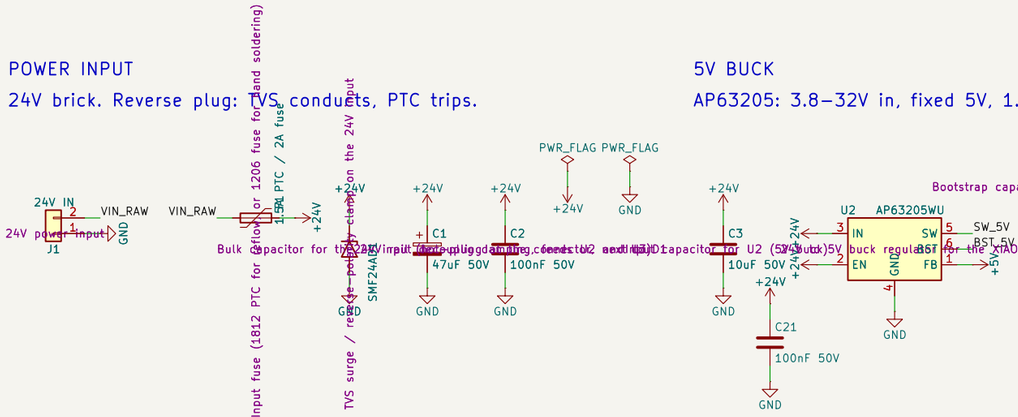
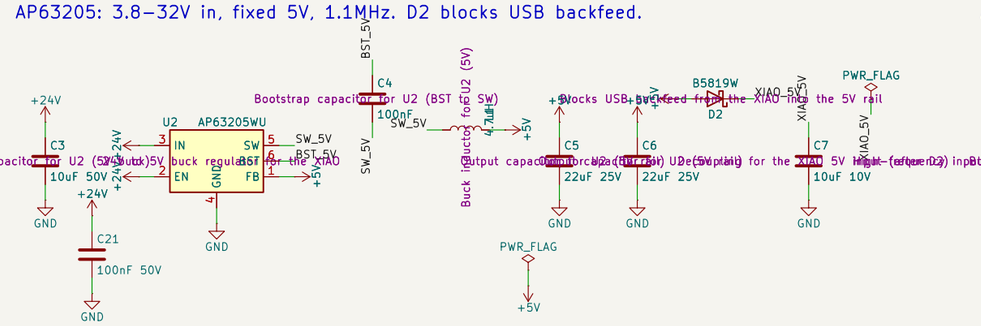
POWER INPUT

24V brick. Reverse plug: TVS conducts, PTC trips.



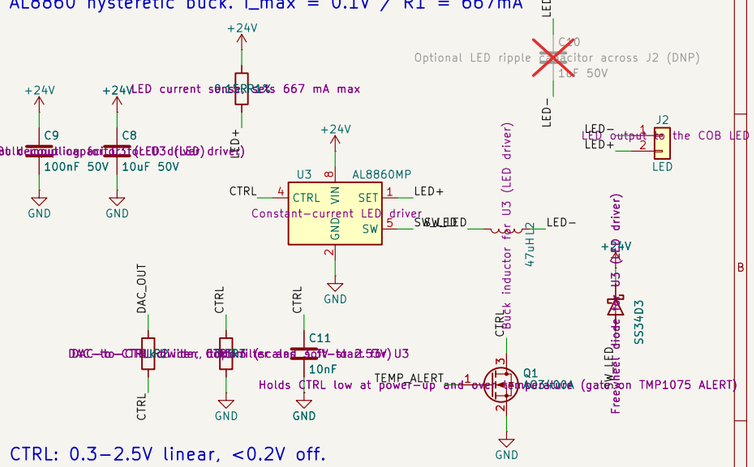
5V BUCK

AP63205: 3.8–32V in, fixed 5V, 1.1MHz. D2 blocks USB backfeed.



LED DRIVER

AL8860 hysteretic buck. I_max = 0.1V / R1 = 667mA



CTRL: 0.3–2.5V linear, <0.2V off.

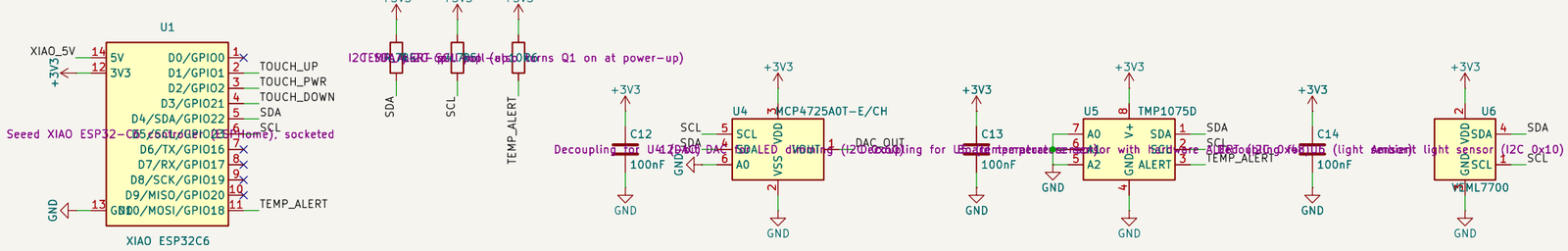
DAC 0–3.3V x 3.3k/4.3k = 0–2.53V (~3600 usable steps).

Power-up: MCP4725 ships at mid-scale, so Q1 holds CTRL low until ESPHome sets

MCU: Seeed XIAO ESP32-C6 (socketed, QT Py footprint)

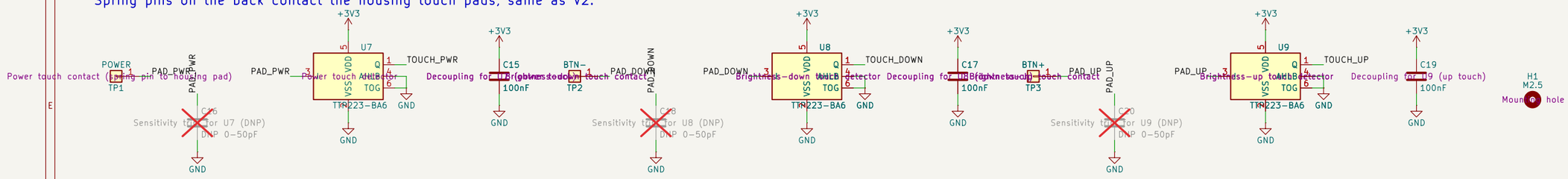
D6/TX (GPIO16) left free: ROM prints on it at boot.

I2C: MCP4725 0x60, TMP1075 0x48, VEML7700 0x10



TOUCH: ESP32-C6 has no touch peripheral -> TTP223 (active-high, direct mode)

Spring pins on the back contact the housing touch pads, same as V2.



ESPHome / Home Assistant native: MCP4725 + AL8860 + AP63205 + TTP223
Derived from Open Task Light Control Board V2 by Steven Bennett

Sheet: /
File: OTL-Control-Board.kicad_sch

Title: Open Task Light Control Board – XIAO ESP32-C6

Size: A3 | Date: 2026-10-08 | **Rev: 3.0**
KiCad E.D.A. 10.0.3 | Id: 1/1